



SHAIK MOHAMMED ZAHEER <smohammedzaheer032@gmail.com>

MERN TASK

1 message

vraj patel <vraj.p@technoyuga.com>

Mon, Jun 1, 2026 at 10:39 AM

To: "smohammedzaheer032@gmail.com" <smohammedzaheer032@gmail.com>

Cc: Technoyuga HR <hr@technoyuga.com>

MERN Stack Evaluation Task (Beginner to Intermediate)

Objective

Build a simple **E-Commerce Management System** with separate **Admin** and **User** roles.

The goal is to evaluate:

- React fundamentals
- Node.js & Express APIs
- MongoDB database design
- Authentication & Authorization
- CRUD operations
- State management
- Code quality and project structure

Tech Stack

Frontend

- React.js
- React Router
- Axios
- Context API (or Redux if preferred)

Backend

- Node.js
- Express.js
- MongoDB
- Mongoose

Authentication

- JWT Authentication
- Password Hashing using bcrypt

User Roles

1. Admin

Admin can:

- Login
 - View Dashboard
 - Manage Products
 - Manage Users
 - View Orders
-

2. User

User can:

- Register
 - Login
 - Forgot Password
 - Reset Password
 - Manage Profile
 - Browse Products
 - Add Products to Cart
 - Place Orders
 - View Order History
-

Database Collections

Users Collection

```
{
  _id,
  firstName,
  lastName,
  email,
  password,
  phone,
  role, // admin or user
  isActive,
  createdAt
}
```

Products Collection

```
{
  _id,
  name,
  description,
  price,
  quantity,
  category,
  image,
}
```

```
    status,  
    createdAt  
  }  
}
```

Cart Collection

```
{  
  _id,  
  userId,  
  items:[  
    {  
      productId,  
      quantity,  
      price  
    }  
  ]  
}
```

Orders Collection

```
{  
  _id,  
  userId,  
  items:[],  
  totalAmount,  
  status,  
  createdAt  
}
```

Authentication Module

User Registration

Fields:

- First Name
- Last Name
- Email
- Password
- Confirm Password

Validations:

- Email unique
 - Password minimum 8 characters
-

Login

Fields:

- Email
- Password

Requirements:

- JWT generation
 - Token expiration
-

Forgot Password

Flow:

1. Enter Email
 2. Generate reset token
 3. Save token in DB
 4. Simulate email sending (console log is acceptable)
 5. Reset password using token
-

Protected Routes

Only authenticated users can access:

- Profile
- Cart
- Orders

Admin-only routes:

- Product Management
 - User Management
-

Admin Panel

Dashboard

Show:

- Total Users
 - Total Products
 - Total Orders
-

User Management

Admin should be able to:

List Users

Columns:

- Name
- Email
- Role
- Status

Actions

- View User
- Activate/Deactivate User
- Delete User

Product Management

Create Product

Fields:

- Product Name
- Description
- Price
- Stock Quantity
- Category
- Image URL

Product Listing

Features:

- Pagination
- Search
- Sorting

Edit Product

Update all product details.

Delete Product

Soft delete preferred.

User Panel

Home Page

Display:

- Product Grid

Each Product Card:

- Image
 - Name
 - Price
 - Add To Cart Button
-

Product Search

Search by:

- Product Name
 - Category
-

Product Filtering

Filter by:

- Category
-

Product Details Page

Display:

- Product Information
 - Stock Availability
 - Add To Cart
-

Cart Module

Add To Cart

Requirements:

- Prevent duplicate entries
 - Increase quantity if already exists
-

Cart Page

Display:

- Product Name
- Price
- Quantity
- Subtotal

Actions:

- Increase Quantity
 - Decrease Quantity
 - Remove Item
-

Cart Summary

Show:

- Total Items
 - Total Amount
-

Order Module

Checkout

Requirements:

- Create Order
- Clear Cart

Order History

User can view:

- Order ID
 - Date
 - Amount
 - Status
-

Order Details

Display:

- Products
 - Quantity
 - Price
-

API Requirements

Authentication APIs

POST /api/auth/register
POST /api/auth/login
POST /api/auth/forgot-password
POST /api/auth/reset-password

User APIs

GET /api/users
GET /api/users/:id
PUT /api/users/:id
DELETE /api/users/:id

Product APIs

POST /api/products
GET /api/products
GET /api/products/:id
PUT /api/products/:id
DELETE /api/products/:id

Cart APIs

POST /api/cart/add
GET /api/cart
PUT /api/cart/update
DELETE /api/cart/remove/:id

Order APIs

POST /api/orders
GET /api/orders
GET /api/orders/:id

Additional Requirements

Backend

- MVC architecture
 - Proper validation
 - Error handling middleware
 - Environment variables
 - Role-based authorization middleware
-

Frontend

- Responsive UI
 - Form validations
 - Loading states
 - Error messages
 - Protected routes
-

Bonus Features (Optional)

1. Product image upload using Cloudinary.
 2. Refresh token implementation.
 3. Wishlist functionality.
 4. Admin order status update.
 5. Product reviews and ratings.
 6. Docker setup.
 7. Unit tests.
-

Evaluation Criteria (100 Marks)

Area	Marks
Authentication & Authorization	20
Backend APIs	20
MongoDB Design	10
React Frontend	20
Cart & Order Flow	10
Validation & Error Handling	10
Code Structure & Best Practices	10

Expected Completion Time

- Beginner: 3–5 days
- Intermediate: 1–2 days

This task is comprehensive enough to assess whether a candidate can independently build and maintain a real-world MERN application while following common development practices.

--

Thanks & Regards,

Vraj Patel
MERN Stack Developer
Technoyuga Soft Pvt Ltd.
vraj.p@technoyuga.com